RealMan Robot rm_driver User Manual V1.7

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Revision History

No.	Date	Comment
V1.0	2024-2-7	Draft
V1.1	2024-7-8	Amend(Add GEN72 adapter files)
V1.2	2024-9-10	Amend(Add ECO63 adapter files)
V1.2.1	2024-10-31	Amend(Add dexterous hand high-speed adaptation)
V1.3	2024-12-25	Amend(Add UDP reporting adaptation)
V1.4	2024-12-25	Amend(Add UDP reporting adaptation)
V1.5	2025-5-29	Amend(Adapt to fourth-generation controllers, add a version query interface, Cartesian space linear offset motion interface, Modbus interface, and trajectory list interface; see the topic interface description document for details)
V1.6	2025-11-13	Amend(Add basic UDP enable configurations)
V1.7	2026-4-16	Amend(Add ECO62 and RX75 adapter files)

Content

1. rm_driver package description

rm_driver package is very important in the ROS2 robotic arm package. This package realizes the function of controlling the robotic arm through communication between ROS and the robotic arm. The package will be introduced in detail in the following text through the following aspects:

- 1.Package use.
- 2.Package architecture description.
- 3.Package topic description.

Through the introduction of the three parts, it can help you:

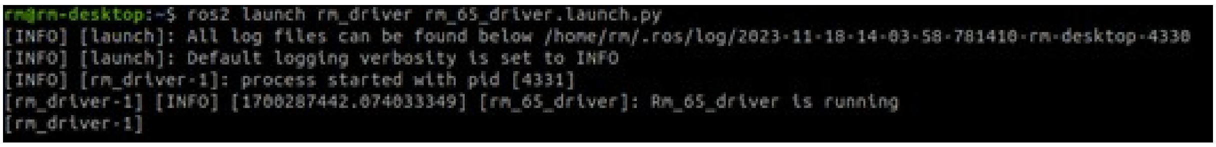
- 1.Understand the package use.
- 2.Familiar with the file structure and function of the package.
- 3.Familiar with the topic related to the package for easy development and use.

Source code address:https://github.com/RealManRobot/ros2_rm_robot.git.

2. rm_driver package use

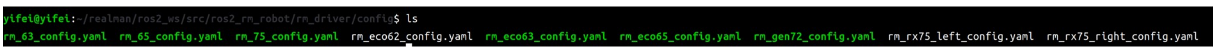
2.1 Basic use of the package

First, after configuring the environment and completing the connection, we can directly start the node and control the robotic arm through the following command. The current control is based on the fact that we have not changed the IP of the robotic arm, which is still 192.168.1.18. rm@rm-desktop:~$ ros2 launch rm_driver rm__driver.launch.py In practice, the above needs to be replaced by the actual model of the robotic arm. The available models of the robotic arm are 65, 63, eco65, eco63, eco62, gen72, 75, and rx75. The following screen will appear if the underlying driver is successfully started.



2.2 Advanced use of the package

When our robotic arm's IP is changed, our start command is invalid. If we use the above command directly, we cannot successfully connect to the robotic arm. We can re-establish the connection by modifying the following configuration file. The configuration file is located in the config folder under our rm_driver package.



The contents of the configuration file are as follows:

```
rm_driver:
  ros__parameters:
    #robot param
    arm_ip: "192.168.1.18"          # Set the IP address for the TCP connection
    tcp_port: 8080#                 # Set the port for the TCP connection

    arm_type: "RM_65"               # set the robotic arm model
    arm_dof: 6                      # Set the degree of freedom of the robotic arm
```

```
    udp_ip: "192.168.1.10"        # set the udp active reporting IP address
    udp_cycle: 5                  # the active reporting cycle of UDP, which needs to
be a multiple of 5.
    udp_port: 8089                # Set the udp active reporting port
    udp_force_coordinate: 0       # Set the base coordinate of the six-axis force when
the system is forced, where 0 is the sensor coordinate system, 1 is the current work
coordinate system, and 2 is the current tool coordinate system
    udp_hand: false               # Set the udp hand reporting enable
    udp_plus_base: false          # Set the udp plus base reporting enable
    udp_plus_state: false         # Set the udp plus state reporting enable
    udp_joint_speed_state: true   # Set the udp joint speed reporting enable
    udp_lift_state: true          # Set the udp lift state reporting enable
    udp_expand_state: false       # Set the udp expand state reporting enable
    udp_arm_current_status: true  # Set the udp arm_current status reporting enable
    udp_aloha_state: true         # Set the udp plus base reporting enable
    trajectory_mode: 0            #When the high following mode is set, multiple modes
are supported, including 0- complete transparent transmission mode, 1- curve fitting
mode and 2- filtering mode.
    radio: 0                    #Set the smoothing coefficient in curve fitting mode
(range 0-100) or the filter parameter in filtering mode (range 0-1000). The higher the
value, the better the smoothing effect.
    arm_joints: ["joint1", "joint2", "joint3", "joint4", "joint5", "joint6"]
```

There are mainly the following parameters.

- arm_ip: This parameter represents the current IP of the robotic arm
- tcp_port: set the port when TCP is connected.
- arm_type: This parameter represents the current model of the robotic arm. The parameters that can be selected are RM_65 (65 series), RM_eco65 (ECO65 series), RM_eco63 (ECO63 series), RM_eco62 (ECO62 series), RM_63 (63 series), GEN_72 (GEN72 series), and RM_75 (75 series; the RX75 dual-arm configuration also uses this model type for each arm).
- arm_dof: set the degree of freedom of the robotic arm. 6 is 6 degrees of freedom, and 7 is 7 degrees of freedom.
- udp_ip: set the udp active reporting IP address.
- udp_cycle: the active reporting cycle of UDP, which needs to be a multiple of 5.
- udp_port: set the udp active reporting port.
- udp_force_coordinate: set the base coordinate of the six-axis force when the system is forced, where 0 is the sensor coordinate system (original data), 1 is the current work coordinate system, and 2 is the current tool coordinate system.
- udp_hand: false # Set the udp hand reporting enable
- udp_plus_base: false # Set the udp plus base reporting enable
- udp_plus_state: false # Set the udp plus state reporting enable
- udp_joint_speed_state: true # Set the udp joint speed reporting enable
- udp_lift_state: true # Set the udp lift state reporting enable
- udp_expand_state: false # Set the udp expand state reporting enable
- udp_arm_current_status: true # Set the udp arm_current status reporting enable
- udp_aloha_state: true # Set the udp plus base reporting enable
- trajectory_mode：When the high following mode is set, multiple modes are supported, including 0- complete transparent transmission mode, 1- curve fitting mode and 2- filtering mode.
- radio：Set the smoothing coefficient in curve fitting mode (range 0-100) or the filter parameter in filtering mode (range 0-1000). The higher the value, the better the smoothing effect.
- In practice, we choose the corresponding launch file to start, which will automatically select the correct model. If there are special requirements, you can modify the

corresponding parameters here. After modification, recompile the configuration in the workspace directory, and then the modified configuration will take effect.

Run the colcon build command in the workspace directory.

```
rm@rm-desktop: ~/ros2_ws$ colcon build
```

After successful compilation, follow the above commands to start the package.

3. rm_driver package architecture description

3.1 Overview of package files

The current rm_driver package is composed of the following files.

```
├── CMakeLists.txt                # compilation rule file
├── config                        # config folder
│   ├── rm_63_config.yaml         # 63 configuration file
│   ├── rm_65_config.yaml         # 65 configuration file
│   ├── rm_75_config.yaml         # 75 configuration file
│   ├── rm_eco62_config.yaml      # eco62 configuration file
│   ├── rm_eco65_config.yaml      # eco65 configuration file
│   ├── rm_eco63_config.yaml      # eco63 configuration file
│   ├── rm_gen72_config.yaml      # gen72 configuration file
│   ├── rm_rx75_left_config.yaml  # RX75 left-arm configuration file
│   └── rm_rx75_right_config.yaml # RX75 right-arm configuration file
├── doc
│   ├── RealMan Robotic Arm rm_driver Topic Detailed Description (ROS2).md
│   ├── rm_driver1.png
│   ├── rm_driver2.png
│   ├── rm_driver3.png
│   ├── rm_driver4.png
│   └── 睿尔曼机械臂ROS2rm_driver话题详细说明.md
├── include                       # dependency header file folder
│   └── rm_driver
│       ├── cJSON.h                # API header file
│       ├── constant_define.h      # API header file
│       ├── rman_int.h             # API header file
│       ├── rm_base_global.h       # API header file
│       ├── rm_base.h              # API header file
│       ├── rm_define.h            # API header file
│       ├── rm_driver.h            # rm_driver.cpp header file
│       ├── rm_praser_data.h       # API header file
│       ├── rm_queue.h             # API header file
│       ├── rm_service_global.h    # API header file
│       ├── rm_service.h           # API header file
│       └── robot_define.h         # API header file
├── launch
│   ├── rm_63_driver.launch.py     # 63 launch file
│   ├── rm_65_driver.launch.py     # 65 launch file
│   ├── rm_75_driver.launch.py     # 75 launch file
│   ├── rm_eco62_driver.launch.py  # eco62 launch file
│   ├── rm_eco65_driver.launch.py  # eco65 launch file
│   ├── rm_eco63_driver.launch.py  # eco63 launch file
│   ├── rm_gen72_driver.launch.py  # gen72 launch file
│   └── rm_rx75_driver.launch.py   # RX75 launch file
├── lib
│   ├── libRM_Service.so -> libRM_Service.so.1.0.0         # API library file
│   ├── libRM_Service.so.1 -> libRM_Service.so.1.0.0       # API library file
│   ├── libRM_Service.so.1.0 -> libRM_Service.so.1.0.0     # API library file
│   ├── libRM_Service.so.1.0.0                             # API library file
│   ├── linux_arm_service_release_v4.3.7.t7.tar.bz2        # API library file
│   └── linux_x86_service_release_v4.3.7.t7.tar.bz2        # API library file
├── package.xml                                            # dependency declaration file
├── README_CN.md
├── README.md
└── src
    └── rm_driver.cpp                                      # driver code source file
```

4. rm_driver topic description

rm_driver has many topics, and you can learn about the topic information through the following commands.

```
rm@rm-desktop:~$ ros2 topic list
/joint_states
/parameter_events
/rm_driver/change_work_frame_cmd
/rm_driver/change_work_frame_result
/rm_driver/clear_force_data_cmd
/rm_driver/clear_force_data_result
/rm_driver/force_position_move_joint_cmd
/rm_driver/force_position_move_pose_cmd
/rm_driver/get_all_tool_frame_cmd
/rm_driver/get_all_tool_frame_result
/rm_driver/get_all_work_frame_cmd
/rm_driver/get_all_work_frame_result
/rm_driver/get_arm_software_version_cmd
/rm_driver/get_arm_software_version_result
/rm_driver/get_curr_workFrame_cmd
/rm_driver/get_curr_workFrame_result
/rm_driver/get_current_arm_original_state_result
/rm_driver/get_current_arm_state_cmd
/rm_driver/get_current_arm_state_result
/rm_driver/get_current_tool_frame_cmd
/rm_driver/get_current_tool_frame_result
/rm_driver/get_lift_state_cmd
/rm_driver/get_lift_state_result
/rm_driver/get_realtime_push_cmd
/rm_driver/get_realtime_push_result
/rm_driver/move_stop_cmd
/rm_driver/move_stop_result
/rm_driver/movec_cmd
/rm_driver/movec_result
/rm_driver/movej_canfd_cmd
/rm_driver/movej_cmd
/rm_driver/movej_p_cmd
/rm_driver/movej_p_result
/rm_driver/movej_result
/rm_driver/movel_cmd
/rm_driver/movel_result
/rm_driver/movep_canfd_cmd
/rm_driver/set_force_postion_cmd
/rm_driver/set_force_postion_result
/rm_driver/set_gripper_pick_cmd
/rm_driver/set_gripper_pick_on_cmd
/rm_driver/set_gripper_pick_on_result
/rm_driver/set_gripper_pick_result
/rm_driver/set_gripper_position_cmd
/rm_driver/set_gripper_position_result
```

```
/rm_driver/set_hand_angle_cmd
/rm_driver/set_hand_angle_result
/rm_driver/set_hand_force_cmd
/rm_driver/set_hand_force_result
/rm_driver/set_hand_posture_cmd
/rm_driver/set_hand_posture_result
/rm_driver/set_hand_seq_cmd
/rm_driver/set_hand_seq_result
/rm_driver/set_hand_speed_cmd
/rm_driver/set_hand_speed_result
/rm_driver/set_joint_err_clear_cmd
/rm_driver/set_joint_err_clear_result
/rm_driver/set_lift_height_cmd
/rm_driver/set_lift_height_result
/rm_driver/set_lift_speed_cmd
/rm_driver/set_lift_speed_result
/rm_driver/set_realtime_push_cmd
/rm_driver/set_realtime_push_result
/rm_driver/set_tool_voltage_cmd
/rm_driver/set_tool_voltage_result
/rm_driver/start_force_position_move_cmd
/rm_driver/start_force_position_move_result
/rm_driver/stop_force_position_move_cmd
/rm_driver/stop_force_position_move_result
/rm_driver/stop_force_postion_cmd
/rm_driver/stop_force_postion_result
/rm_driver/udp_arm_coordinate
/rm_driver/udp_arm_err
/rm_driver/udp_arm_position
/rm_driver/udp_joint_error_code
/rm_driver/udp_one_force
/rm_driver/udp_one_zero_force
/rm_driver/udp_six_force
/rm_driver/udp_six_zero_force
/rm_driver/udp_sys_err
/rosout
```

It is mainly for the application of API to achieve some of the robotic arm functions; for a more complete introduction and use, please see the special document "RealMan Robotic Arm ROS2 Topic Detailed Description (https://github.com/RealManRobot/ros2_rm_robot/blob/humble/rm_driver/doc/RealMan%20Robotic%20Arm%20rm_driver%20Topic%20Detailed%20Description%20(ROS2).md)".